

Two-Stage Portfolio Optimization for Taiwan Equities with CVaR and Black-Swan Risk Controls

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1 Overview and Notation

In the Taiwan stock market, where information flows rapidly and intraday volatility is high, many investors—lacking systematic analysis—are easily influenced by emotions, leading to irrational trading decisions.

Our proposed approach automatically retrieves all market data from the previous trading day (including prices, volumes, technical indicators, and brokerage branch activity) one hour before the market opens each day. Using a 60-day rolling window, the system records and analyzes each stock’s **momentum** and **trend**.

Next, through a risk-control and optimization algorithm, we incorporate the following constraints:

- **Maximum brokerage effect:** Avoid over-concentrating capital in stocks dominated by a single brokerage to reduce liquidity risk.
- **Deviation rate limit:** Prevent purchases at excessively extended price levels far from the average, which carry higher risk.
- **Risk matrix:** Assess correlations and volatilities among stocks to ensure that the overall portfolio remains stable.

On this foundation, we interpret the market’s “correction energy” (i.e., the potential for short-term mean reversion or trend continuation) as a **high-energy signal**. We then apply a **Simulated Annealing** algorithm to determine the optimal portfolio such that:

- **Daily return** is maintained at approximately 1%.
- **Daily risk (volatility)** is controlled at approximately 1%.

With this strategy, the program calculates the day’s investment allocations and buy/sell limits before the market opens, then automatically executes trades during the session. In other words, investors do not need to monitor the market throughout the day—an algorithm can maintain both returns and risks within the desired target ranges.

We produce orders after close on day t and execute at the open on $t+1$. Stage 1 generates *continuous* candidate portfolios via a convex Second-Order Cone Programing (SOCP) with linear and impact costs and a volatility cap scaled by a black-swan multiplier $m \in (0, 1]$. A policy layer then selects among candidates by maximizing long-horizon right-tail CVaR. Stage 2 (DSA) maps the chosen weight vector to *integer shares* with exact Taiwan fees/taxes/impact.

Vectors are column vectors. For weights $w \in \mathbb{R}^N$, $\mathbf{1}^\top w = 1$. The risk matrix is $\hat{\Sigma}$. Capital is P_0 . Prices p_i at close t imply $\omega_i(n) = n_i p_i / P_0$.

2 Daily Close Data Input Specification

Next, we will define all names directly using common stock names. All data are aligned to the t close; orders are for $t+1$.

2.1 1) Price and Volume (Raw Fields)

Frequency: daily operation. **Timing:** Calculate at 6 a.m. and place orders at the market open. Per security $(i, t+1)$:

- O, H, L, C (close adjusted for splits/dividends), V (shares), VAL (TWD).
- `inside_vol`, `outside_vol` (optional; order counts may substitute).

- broker_netbuy_topK, plus broker_buy_topK/broker_sell_topK.
- lot_size, odd_lot_unit, industry/group, tradable_flag.

Cleaning:

Data Cleaning Rules

Objective. Ensure that historical market data are clean, continuous, and usable so that outliers or gaps do not distort model calculations.

1. **Price Adjustment (Corporate Actions).** Adjust $O/H/L/C$ (open, high, low, close) to reflect stock splits, dividends, and other corporate actions, ensuring price continuity. Do *not* adjust VAL (traded value), as it represents actual transaction amounts.
2. **Missing-Value Imputation.** When missing observations occur, apply forward-fill or back-fill to preserve time-series continuity.
3. **Non-tradable Flagging.** If volume (V) or traded value (VAL) equals zero, mark the security as non-tradable (suspension or insufficient liquidity).
4. **Winsorization of Extremes.** Truncate extreme movements beyond $\pm 8\sigma$ by winsorizing to the 1st/99th percentiles, mitigating the impact of outliers and misprints on distributional statistics.

2.2 2) Features

2.1 Price/Momentum

These formulas belong to the category of **Price and Momentum features** and the **Garman–Klass volatility estimator**. In quantitative investment models (especially for backtesting, forecasting, and portfolio optimization), they serve as **pre-processing feature engineering** steps.

First, the **Gap** measures the overnight price jump from the previous day’s close to the current day’s open:

$$\text{Gap}_{i,t} = \frac{O_{i,t} - C_{i,t-1}}{C_{i,t-1}}$$

It reflects overnight market sentiment and news impact, making it an important signal for next-day and intraday strategies.

Momentum captures price trends over different time horizons:

$$\text{Mom}_{i,t}^{(k)} = \frac{C_{i,t} - C_{i,t-k}}{C_{i,t-k}}, \quad k \in \{5, 10, 20\}$$

It serves as a key indicator for identifying short- to medium-term market direction.

The **Garman–Klass volatility estimator** is given by:

$$\sigma_{\text{GK},i,t}^2 = \frac{1}{2} \left(\ln \frac{H_{i,t}}{L_{i,t}} \right)^2 - (2 \ln 2 - 1) \left(\ln \frac{C_{i,t}}{O_{i,t}} \right)^2$$

By combining the high $H_{i,t}$, low $L_{i,t}$, open $O_{i,t}$, and close $C_{i,t}$ prices, it estimates volatility more accurately than the simple close-to-close method and reduces noise.

The objectives of computing these indicators are:

1. **Remove the effects of price units and scale**, ensuring comparability across different stocks.

2. **Capture structural market features** (gaps, trends, volatility).
3. **Provide standardized inputs for risk control and portfolio optimization** (e.g., CVaR, SOCP, QUBO, simulated annealing).

In other words, this step transforms raw market data into standardized, robust, and information-rich numerical features that can be directly fed into optimizers for return and risk calculations.

2.2 Volume/Liquidity

This set of formulas represents two market activity features:

1. Liquidity

$$\text{Liquidity}_{i,t} = \log(\text{VAL}_{i,t} + 1)$$

- $\text{VAL}_{i,t}$ denotes the traded value (trade volume $\times$ trade price) of stock i at time t .
- The $+1$ term prevents mathematical errors when the traded value is zero.
- Taking the natural logarithm compresses the range of traded values, making liquidity more comparable across different stocks.

2. Turnover Share

$$\text{TurnoverShare}_{i,t} = \frac{\text{VAL}_{i,t}}{\sum_j \text{VAL}_{j,t}}$$

- The numerator is the traded value of stock i at time t .
- The denominator is the total traded value of all stocks in the same period.
- This ratio represents the proportion of market turnover contributed by stock i , which can be interpreted as its *market turnover share*, reflecting trading activity level.

2.3 Order Imbalance

Order Imbalance is commonly measured as:

$$\text{Imb}_{i,t} = \frac{\text{inside_vol} - \text{outside_vol}}{\text{inside_vol} + \text{outside_vol} + 1}$$

where inside_vol denotes the traded or quoted volume at the best bid/ask (inside quote), and outside_vol denotes the traded or quoted volume away from the best bid/ask (outside quote). The $+1$ in the denominator ensures numerical stability by avoiding division by zero. Values of Imb close to $+1$ indicate that buy (or sell) orders are heavily concentrated near the best quote, while values close to -1 reflect stronger opposite-side pressure with more orders placed away from the best quote. Values near 0 imply a relatively balanced order book. This feature can be computed from Limit Order Book (LOB) data in real time and is often used as an input for assessing short-term liquidity and market pressure.

2.4 Deviation (BIAS)

Deviation (BIAS) measures the extent to which a stock price deviates from its moving average:

$$\text{BIAS}_{i,t} = \frac{C_{i,t} - MA_n(C_{i,\cdot})}{MA_n(C_{i,\cdot})}, \quad n = 10 \quad (\text{or EMA with half-life} = 10)$$

where $C_{i,t}$ is the closing price of stock i at time t , and MA_n denotes the n -day moving average (or an exponentially weighted moving average, EMA, with a half-life of 10). The BIAS indicator helps identify whether a stock is excessively deviating from its average price level, serving as a tool to support trading decisions.

Broker Strength (BF) quantifies the proportion of net buying by selected brokers relative to the total traded value:

$$\text{BF}_{i,t} = \frac{\sum_{k=1}^K \text{NetBuy}_{i,t}^{(k)}}{\text{VAL}_{i,t} + 1} \quad (\text{EMA over 3-5 days})$$

Here, $\text{NetBuy}_{i,t}^{(k)}$ is the net buy amount of stock i by broker k at time t , K is the number of brokers included in the calculation, and $\text{VAL}_{i,t}$ is the traded value on day t . The +1 in the denominator avoids division-by-zero errors. A 3-5 day EMA smoothing can be applied to BF to observe short-term concentration of institutional or broker-driven capital flows.

2.6 Processing

Daily cross-sectional z-score or quantile map to $[-3, 3]$; fill missing with 0 or industry median; winsorize 1%/99%.

2.3 3) Risk Matrix $\hat{\Sigma}$

EWMA with half-life h (default 12): $\lambda = e^{-\ln 2/h}$,

$$\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda)(r_t - \bar{r})(r_t - \bar{r})^\top,$$

then Ledoit-Wolf shrinkage $\hat{\Sigma} = \rho \text{diag}(\Sigma_t) + (1 - \rho)\Sigma_t$, $\rho \in [0.1, 0.4]$. Optional factor denoising. Per-asset daily σ_i from 30-day log-return std or GK.

2.4 4) Cost Model (Taiwan fees + impact)

For $A_i = |\Delta n_i| p_i$ TWD (or $A_i = P_0 |\Delta w_i|$):

$$\text{Fee}_{i,\text{side}} = \max(20, 0.001425 A_i) \mathbf{1}_{A_i > 0}, \quad \text{Tax}_i = 0.003 A_i \mathbf{1}_{\text{sell}}, \quad \text{Impact}_i = \gamma_i \sigma_i \sqrt{\frac{A_i}{\text{ADV}_i}} A_i,$$

$\text{ADV}_i = \frac{1}{30} \sum_{s=t-29}^t \text{VAL}_{i,s}$, γ_i init 4×10^{-4} (refit monthly). Total $T_c = \sum_i (\text{Fee}_i^{\text{buy}} + \text{Fee}_i^{\text{sell}} + \text{Tax}_i + \text{Impact}_i)$. Stage 1 uses linear+impact only; NT\$20/no-trade handled in Stage 2.

2.5 5) Black-Swan Scaling m

Triggers (examples): index return $\leq -3\% \Rightarrow m = 0.7$; 30D vol jump ratio $> 1.5 \Rightarrow m = 0.8$; VIX-proxy $> 2 \times \text{mean} \Rightarrow m = 0.7$; event list $\Rightarrow m = 0.5$. Risk cap becomes $\sqrt{w^\top \hat{\Sigma} w} \leq m \sigma_{\text{target}}$ with $\sigma_{\text{target}} = 1\%$.

2.6 6) Calibrated Forecast for Stage 1

With quantile calibration $g(\hat{r})$:

$$\hat{r}_i^{\text{enh}} = g(\hat{r}_i) + \theta_b \text{BF}_i - \kappa_{\text{bias}} \max\{0, |\text{BIAS}_i| - \theta_{\text{bias}}\},$$

$\theta_b \in [0.1, 0.5]$, $\theta_{\text{bias}} = 10\%$, $\kappa_{\text{bias}} \in [0.5, 1.5]$.

2.7 7) Minimal Field Table

Field	Type/Unit	Description
date	date	Trading day (Taipei TZ)
sid	string	Stock code
O,H,L,C	TWD	Adjusted prices
V	shares	Volume
VAL	TWD	Traded value
inside_vol/outside_vol	shares	(Optional)
broker_netbuy_topK	TWD/shares	Top- K net buy
group	category	Industry/group
tradable_flag	0/1	Tradable
bias_10	ratio	10D deviation
bf_top5	ratio	Broker strength
mom_5,10,20	ratio	Momentum
gap, imb, sigma_gk	ratio	Gap/Imb/GK vol
adv_30, sigma_30	TWD/ratio	ADV/30D vol
rhat_raw, rhat_enh	ratio	Forecasts

3 Prediction Layer (Detailed)

3.1 Target Definition

Choose one: OO, OC, or CC; fix throughout. Example:

$$r_{i,t+1}^{OO} = \frac{O_{i,t+1} - O_{i,t}}{O_{i,t}}, \quad r_{i,t+1}^{OC} = \frac{C_{i,t+1} - O_{i,t+1}}{O_{i,t+1}}, \quad r_{i,t+1}^{CC} = \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}}.$$

3.2 Walk-Forward Training

Train on $[t - W, t - 1]$; predict on t . Ridge/Lasso/GBDT with time weights $\omega_s = e^{-(t-1-s)/\tau}$, τ HL 60:

$$\min_{\beta} \sum_{s=t-W}^{t-1} \sum_i \omega_s (r_{i,s+1} - \beta^\top x_{i,s})^2 + \lambda \|\beta\|_q, \quad q \in \{1, 2\}.$$

Standardize features cross-sectionally; keep hyperparameters fixed 6–12 months; restrict to tradable names.

3.3 Quantile Calibration and Enhancement

ECDF $\rightarrow$ probit $z = \Phi^{-1}(u)$; align to $q_{50} = 0.3\%$, $q_{75} = 1\%$ via $g(\hat{r}) = az + b$ with $a = (q_{75} - q_{50})/0.67449$, $b = q_{50}$. Then apply BF/BIAS adjustment as above. Avoid leakage (use $t - 1$ broker data if $T+1$ availability).

4 Candidate Portfolios via SOCP

For each parameter setting, solve

$$\begin{aligned} & \max_w \hat{r}^{\text{enh}}^\top w - c_{\text{lin}}(w) - c_{\text{imp}}(w) \\ & \text{s.t. } \mathbf{1}^\top w = 1, \quad w \geq 0, \quad \|L^\top w\|_2 \leq m\sigma_\star, \quad \sum_{i \in g} w_i \leq \text{cap}_g, \quad \|W(w - w_0)\|_1 \leq \tau_{\text{max}}, \\ & \quad 0 \leq w_i \leq w_{i,\text{base}}^{\text{max}} (1 + a_b \text{BF}_i) (1 - a_{\text{bias}} \max\{0, |\text{BIAS}_i| - \theta_{\text{bias}}\}). \end{aligned} \tag{1}$$

c_{lin} uses buy/sell side coefficients (0.001425 vs 0.004425). Impact: piecewise-linear approx. to $|\Delta w|^{3/2}$ or power-cone. Sweep $\sigma_* \in \{0.6, 0.8, 1.0, 1.2\} \% \times m$, $\tau_{\text{max}} \in \{10, 15, 20, 25\} \%$, impact scales $\{0.7, 1.0, 1.3\}$, small cap jitters $\pm 5\%$. De-duplicate via ℓ_1 radius (e.g., 2%) or k-means; typical $K = 15\text{--}30$.

cvxpy sketch (piecewise impact).

```
w = cp.Variable(N)
z_p = cp.Variable(N, nonneg=True); z_m = cp.Variable(N, nonneg=True)
s = cp.Variable((N, Kseg), nonneg=True)
cons = [cp.sum(w)==1, w>=0, cp.norm(L.T@w,2) <= m*sigma_star]
cons += [w - w0 == z_p - z_m, Wturn @ (z_p + z_m) <= tau_max]
cons += [cp.sum(s, axis=1) == cp.abs(w - w0)]
for g, idx in groups.items(): cons += [cp.sum(w[idx]) <= cap_g[g]]
cons += [w <= caps_i]
clin = k_buy @ z_p + k_sell @ z_m
cimp = cp.sum(cp.multiply(a_scaled, s))
obj = cp.Maximize(r_enh @ w - clin - cimp - eps*cp.sum_squares(w))
cp.Problem(obj, cons).solve(solver=cp.ECOS)
```

5 Policy Layer: Right-Tail Return CVaR (MDP $\rightarrow$ LP)

5.1 Right-Tail CVaR

$$\text{CVaR}_\alpha(\xi) = \min_{y \in [L, U]} \left\{ y + \frac{1}{1-\alpha} \mathbb{E}[(\xi - y)_+] \right\}.$$

5.2 MDP Abstraction

States s (regime buckets); actions a are candidates $w^{(k)}$; reward $R(s, a)$ is next-day *net* return (fees/tax/impact included); transitions $P(s'|s)$ from counts (extend to $P(s'|s, a)$ if needed). For each (s, a) , draw M samples $R^{(\ell)}(s, a)$ (rolling or bootstrap).

5.3 Dual LP on Occupation Measures

Let $x(i, a) \geq 0$ be steady-state frequencies with flow conservation

$$\sum_a x(j, a) - \sum_{i, a} P(j|i, a) x(i, a) = 0, \quad \forall j, \quad \sum_{i, a} x(i, a) = 1.$$

Restrict threshold y to sample supports $R^{(\ell)}(i_0, a_0)$ and solve

$$\begin{aligned} \max_{x, z} \quad & z \\ \text{s.t.} \quad & \frac{1}{M} \sum_{\ell=1}^M \sum_{(i, a)} x(i, a) \left[R^{(\ell)}(i_0, a_0) + \frac{1}{1-\alpha} (R^{(\ell)}(i, a) - R^{(\ell)}(i_0, a_0))_+ \right] \geq z, \\ & \forall (i_0, a_0) \text{ (or a reduced set),} \\ & \sum_a x(j, a) - \sum_{i, a} P(j|i, a) x(i, a) = 0, \quad \forall j, \quad \sum_{i, a} x(i, a) = 1, \quad x(i, a) \geq 0. \end{aligned} \tag{2}$$

Recover policy:

$$\pi(a|s) = \frac{x(s, a)}{\sum_b x(s, b)} \quad (\text{if the denominator is 0, the state is non-recurrent}).$$

Extreme-point structure implies most states pick a single action; a few randomize over two.

Sizing. Vars: $|S||A| + 1$; constraints: $|S| + 1 + K_y$. Use cutting-planes to add violated thresholds. Typical scale: $|S| = 20\text{--}60$, $|A| = 10\text{--}30$, $K_y = 50\text{--}200$ (HiGHS/Gurobi/CPLEX: sub-second to seconds).

6 Execution Layer: Discrete Simulated Annealing (DSA)

6.1 Objective in Share Space

With $n \in \mathbb{Z}_{\geq 0}^N$ and $\omega(n) = n \odot p/P_0$,

$$C(n) = (\hat{r}^{\text{enh}})^\top \omega - \frac{\lambda}{2} \omega^\top \hat{\Sigma} \omega - \frac{T_c(n, n_0)}{P_0} - \lambda_B \left(\sum_i \omega_i - 1 \right)^2.$$

Hard rejections: $\sqrt{\omega^\top \hat{\Sigma} \omega} > m\sigma_\star$, name/group cap breach, per-move bound, or reduce-only violation.

6.2 Initialization

$n_i^{(0)} = \text{round_to_lot}(\frac{w_i^{\text{pol}} P_0}{p_i})$; fix budget to tolerance ε by trimming low-BF/over-deviated names; compute $C(n^{(0)})$.

6.3 Moves and Sampling

Moves: buy/sell by lot, swap(i, j) by lot (budget-neutral), odd-lot ± 1 (late-phase). Sampling: weight names by ADV $_i$; prioritize sells from high-deviation/weak-BF/near-cap, buys from strong-BF/low-deviation; pre-build swap pairs.

6.4 Fast Incremental ΔC

With $\delta = \omega(n') - \omega(n)$ on small set S and $q = \hat{\Sigma} \omega$ cached,

$$\Delta(\text{forecast}) = \sum_{i \in S} \hat{r}_i^{\text{enh}} \delta_i, \quad \Delta(\text{risk}) = \frac{\lambda}{2} (2\delta^\top q + \delta^\top \hat{\Sigma} \delta), \quad \Delta(\text{budget}) = \lambda_B [(s + \Delta s)^2 - s^2].$$

Recompute costs only for changed names; update $q \leftarrow q + \hat{\Sigma} \delta$ on accept.

6.5 Annealing Schedule

Acceptance $p = \min\{1, e^{\Delta C/T}\}$. Auto $T_0 = \eta \cdot \text{median}(|\Delta C|)$ over recent proposals ($\eta \in [1, 3]$); geometric cooling $\rho = 0.97$; 1000–2000 moves/temperature; reheat if no improvement for $L = 5$ temperatures; ramp $\lambda_B : 0.1 \rightarrow 10$. Stop at $T \leq 10^{-4}$ or no gain $< 10^{-6}$ over 5 temperatures or time-to-target.

6.6 Black-Swan Modes

Reduce-only (disable buys) or global scaling $\omega \leftarrow \rho \omega$ ($\rho < 1$) with pre-sell of top- k names to meet $m\sigma_\star$ quickly.

6.7 Outputs

Orders (CSV/DB): sid, side, Delta_n, lot_type, notional, fee, tax, impact, est_slippage_bps. Report: $\sqrt{\omega^\top \hat{\Sigma} \omega}$, group weights, turnover, T_c breakdown, budget error, C , SA telemetry.

7 End-to-End Daily Workflow (Ops)

1. ETL $\rightarrow$ features $\rightarrow \hat{r} \rightarrow g(\hat{r}) \rightarrow \hat{r}^{\text{enh}}$; estimate $\hat{\Sigma}$; compute m .
2. SOCP sweep $\rightarrow K$ candidates $\{w^{(k)}\}$.
3. Build MDP samples $R^{(\ell)}(s, a)$; estimate $P(s'|s)$.
4. Solve dual LP (2) $\rightarrow$ policy $\pi(a|s)$.
5. Observe s_t ; draw $w \sim \pi(\cdot|s_t)$; DSA $\rightarrow$ order list.
6. Risk checks, cost/risk dashboards, TTT and SA telemetry upload; fallback to conservative holdings on failure.

8 Defaults and Retuning Guidelines

$\alpha = 0.95$; $\sigma_\star = 1\%$ (shrink via $m \in [0.5, 0.8]$ on stress); $\tau_{\max} = 20\%$; single-name 20%, Tech 40%; per-move $|\Delta w_i| \leq 3\%$; $\gamma = 4 \times 10^{-4}$ (monthly refit); DSA: $T_0 = 1$, $T_{\min} = 10^{-4}$, cooling 0.97, 1000–2000 moves/temperature, $\lambda_B : 0.1 \rightarrow 10$. Retune: raise α /lower $\sigma_\star, m$ for turbulence; lower $\tau_{\max}$ /raise γ if costs high; widen candidate sweep or jitter caps $\pm 5\%$ for diversity.

9 Deliverables and Interfaces

SOCP Candidate Generator

```
ws = generate_candidates(r_enh, Sigma, w0, caps, groups, adv, gamma,
                        sigma_star=0.01, m=1.0, tau_max=0.2,
                        grid={'sigma':[0.006,0.008,0.010,0.012],
                             'tau':[0.10,0.15,0.20,0.25],
                             'impact':[0.7,1.0,1.3]}, K_target=20)
```

MDP–CVaR Dual-LP Solver

```
pi = solve_policy_dualLP(samples, P, alpha=0.95,
                        critset='quantiles', # or 'cutting_plane'
                        max_cuts=200, solver='highs')
```

DSA Executor

```
orders, report = dsa_exec(w, w0, prices, adv, sigma_i, Sigma,
                        lot_size, odd_lot_unit,
                        fees={'rate':0.001425,'min':20,'tax_sell':0.003},
                        gamma=gamma, m=m,
                        params={'T0':1.0,'Tmin':1e-4,'cool':0.97,
                              'moves_per_T':1500,'lambda_B':[0.1,10]})
```

Appendix: Coherent Risk Measures (Primer)

A coherent risk measure maps loss X to $\rho(X)$ with monotonicity, subadditivity, positive homogeneity, and translation invariance (often also $\rho(0) = 0$). Examples: CVaR/ES and EVaR are

coherent; VaR is not in general; entropic risk is convex but not coherent. Rockafellar–Uryasev representation:

$$\text{CVaR}_\alpha(X) = \min_{\zeta} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[(X - \zeta)_+] \right\}.$$

References

- [1] S. Khan, S. Nazir, and N. Javaid, "On the Maximization of Long-Run Reward CVaR for Markov Decision Processes" arXiv preprint arXiv:2312.01586, 2023. <https://arxiv.org/abs/2312.01586>